In [5]:
```
# Same coop research as research_mcoops.ipynb, for Brooklyn, Queens, Bronx,
REF_YEAR = 2026
OUTER_BOROUGHS = ["Brooklyn", "Queens", "Bronx", "Staten Island"]
TOP_N_NBHD = 25
sns.set_theme(style="whitegrid")


def analyze_borough_coops(borough: str) -> None:
    """EDA + time/place analysis for non-condo coop sales in one borough (ma
    df = borough_frames[borough]
    _bcc = df["BUILDING CLASS CATEGORY"].astype(str).str.upper()
    borough_coops = df[_bcc.str.contains("COOP") & ~_bcc.str.contains("CONDC

    bar = "=" * 72
    print(f"\n{bar}\n {borough.upper()} COOPS \n{bar}")
    print("Total coop sales (excludes CONDO COOPS and similar hybrids):")
    print(borough_coops.shape[0])
    if borough_coops.empty:
        print("No rows — skipping.")
        return

    print("\nBuilding class categories:")
    print(borough_coops["BUILDING CLASS CATEGORY"].unique())
    print("\nColumns:")
    print(borough_coops.columns)
    print("\nHead:")
    print(borough_coops.head())

    all_null = borough_coops.columns[borough_coops.isna().all()]
    print(f"\nEntirely null columns (dropped): {list(all_null)}")
    borough_coops = borough_coops.dropna(axis=1, how="all")
    print(borough_coops.head())

    print("\nFraction null per column:")
    for col in borough_coops.columns:
        print(f"{col}: {borough_coops[col].isna().mean()}")

    borough_coops = borough_coops.drop(columns=["TAX CLASS AT TIME OF SALE"]
    borough_coops["AGE OF THE BUILDING"] = REF_YEAR - borough_coops["YEAR BU
    borough_coops = borough_coops.drop(columns=["BUILDING CLASS CATEGORY"])
    borough_coops = borough_coops.drop(columns=["ADDRESS"])
    print("\nAfter feature tweaks:")
    print(borough_coops.head())

    q1 = borough_coops["SALE PRICE"].quantile(0.25)
    q3 = borough_coops["SALE PRICE"].quantile(0.75)
    iqr = q3 - q1
    upper = q3 + 1.5 * iqr
    mask_out = borough_coops["SALE PRICE"] > upper
    print(f"\nFraction above upper IQR fence: {mask_out.mean()}")
    borough_coops = borough_coops[borough_coops["SALE PRICE"] < upper]
    print(borough_coops.head())

    # --- Time & place (rebuild from raw borough frame so BUILDING CLASS CAT
```

```
    m = borough_frames[borough].copy()
    _bcc2 = m["BUILDING CLASS CATEGORY"].astype(str).str.upper()
    coop_tp = m[_bcc2.str.contains("COOP") & ~_bcc2.str.contains("CONDO")].c

    coop_tp["SALE DATE"] = pd.to_datetime(coop_tp["SALE DATE"])
    bcc_upper = coop_tp["BUILDING CLASS CATEGORY"].astype(str).str.upper()
    coop_tp["COOP_TYPE"] = np.where(
        bcc_upper.str.contains("WALKUP"),
        "Walkup",
        np.where(bcc_upper.str.contains("ELEVATOR"), "Elevator", "Other"),
    )

    coop_tp["YEAR BUILT"] = pd.to_numeric(coop_tp["YEAR BUILT"], errors="coe
    coop_tp["BUILDING_AGE"] = REF_YEAR - coop_tp["YEAR BUILT"]

    q1t = coop_tp["SALE PRICE"].quantile(0.25)
    q3t = coop_tp["SALE PRICE"].quantile(0.75)
    iqrt = q3t - q1t
    upper_fence = q3t + 1.5 * iqrt
    coop_tp = coop_tp[coop_tp["SALE PRICE"] < upper_fence].copy()

    coop_tp["YEAR"] = coop_tp["SALE DATE"].dt.year
    coop_tp["MONTH"] = coop_tp["SALE DATE"].dt.month
    coop_tp["YEAR_MONTH"] = coop_tp["SALE DATE"].dt.to_period("M")
    coop_tp["QUARTER"] = coop_tp["SALE DATE"].dt.to_period("Q")

    print("\nCOOP_TYPE counts:")
    print(coop_tp["COOP_TYPE"].value_counts())
    print(f"\nRows after upper IQR trim: {len(coop_tp):,}")
    print(
        f"Sale date range: {coop_tp['SALE DATE'].min().date()} - {coop_tp['S
    )
    print(f"Neighborhoods: {coop_tp['NEIGHBORHOOD'].nunique()}")

    n_by_nbhd = coop_tp.groupby("NEIGHBORHOOD").size().sort_values(ascending
    top_nbhds = n_by_nbhd.head(TOP_N_NBHD).index

    vol_ym = (
        coop_tp.loc[coop_tp["NEIGHBORHOOD"].isin(top_nbhds)]
        .groupby(["NEIGHBORHOOD", "YEAR_MONTH"], observed=True)
        .size()
        .unstack(fill_value=0)
    )
    vol_ym = vol_ym.loc[n_by_nbhd.head(TOP_N_NBHD).index]
    vol_ym.columns = vol_ym.columns.astype(str)

    fig_w = max(12, vol_ym.shape[1] * 0.35)
    fig, ax = plt.subplots(figsize=(fig_w, 10))
    sns.heatmap(vol_ym, cmap="YlOrRd", ax=ax, linewidths=0, cbar_kws={"label
    ax.set_title(f"{borough} coop sales: top {TOP_N_NBHD} neighborhoods × mo
    ax.set_xlabel("Year-month")
    ax.set_ylabel("Neighborhood")
    plt.tight_layout()
    plt.show()

    peak_month = (
```

```
        coop_tp.loc[coop_tp["NEIGHBORHOOD"].isin(top_nbhds)]
        .groupby("NEIGHBORHOOD", observed=True)["YEAR_MONTH"]
        .agg(lambda s: s.astype(str).value_counts().idxmax())
        .rename("peak_year_month")
    )
    share_peak = (
        coop_tp.loc[coop_tp["NEIGHBORHOOD"].isin(top_nbhds)]
        .groupby(["NEIGHBORHOOD", "YEAR_MONTH"], observed=True)
        .size()
        .groupby(level=0)
        .apply(lambda s: s.max() / s.sum())
        .rename("share_of_nbhd_in_peak_month")
    )
    peak_tbl = pd.concat([peak_month, share_peak], axis=1).sort_values(
        "share_of_nbhd_in_peak_month", ascending=False
    )
    print(
        f"Peak year-month per neighborhood (top {TOP_N_NBHD} by volume) and
    )
    print(peak_tbl.to_string())

    med_all = coop_tp.groupby("YEAR_MONTH", observed=True)["SALE PRICE"].med
    med_type = (
        coop_tp.groupby(["YEAR_MONTH", "COOP_TYPE"], observed=True)["SALE PR
        .median()
        .unstack("COOP_TYPE")
        .sort_index()
    )

    fig, axes = plt.subplots(2, 1, figsize=(12, 8), sharex=True)
    med_all.plot(ax=axes[0], marker="o", legend=False, color="0.2")
    axes[0].set_title(f"Median sale price — all {borough} coops (IQR-trimmed
    axes[0].set_ylabel("Median $")
    axes[0].yaxis.set_major_formatter(
        plt.FuncFormatter(lambda x, _: f"${x/1e6:.2f}M" if x >= 1e6 else f"$
    )

    med_type.plot(ax=axes[1], marker="o")
    axes[1].set_title("Median sale price by month — walkup vs elevator")
    axes[1].set_ylabel("Median $")
    axes[1].yaxis.set_major_formatter(
        plt.FuncFormatter(lambda x, _: f"${x/1e6:.2f}M" if x >= 1e6 else f"$
    )
    axes[1].legend(title="Coop type")
    plt.tight_layout()
    plt.show()

    by_cal_month = (
        coop_tp.groupby("MONTH", observed=True)
        .agg(n=("SALE PRICE", "count"), median_price=("SALE PRICE", "median"
        .sort_index()
    )
    by_cal_month["month_name"] = [
        pd.Timestamp(2020, m, 1).strftime("%B") for m in by_cal_month.index
    ]
    by_cal_month = by_cal_month.set_index("month_name")
```

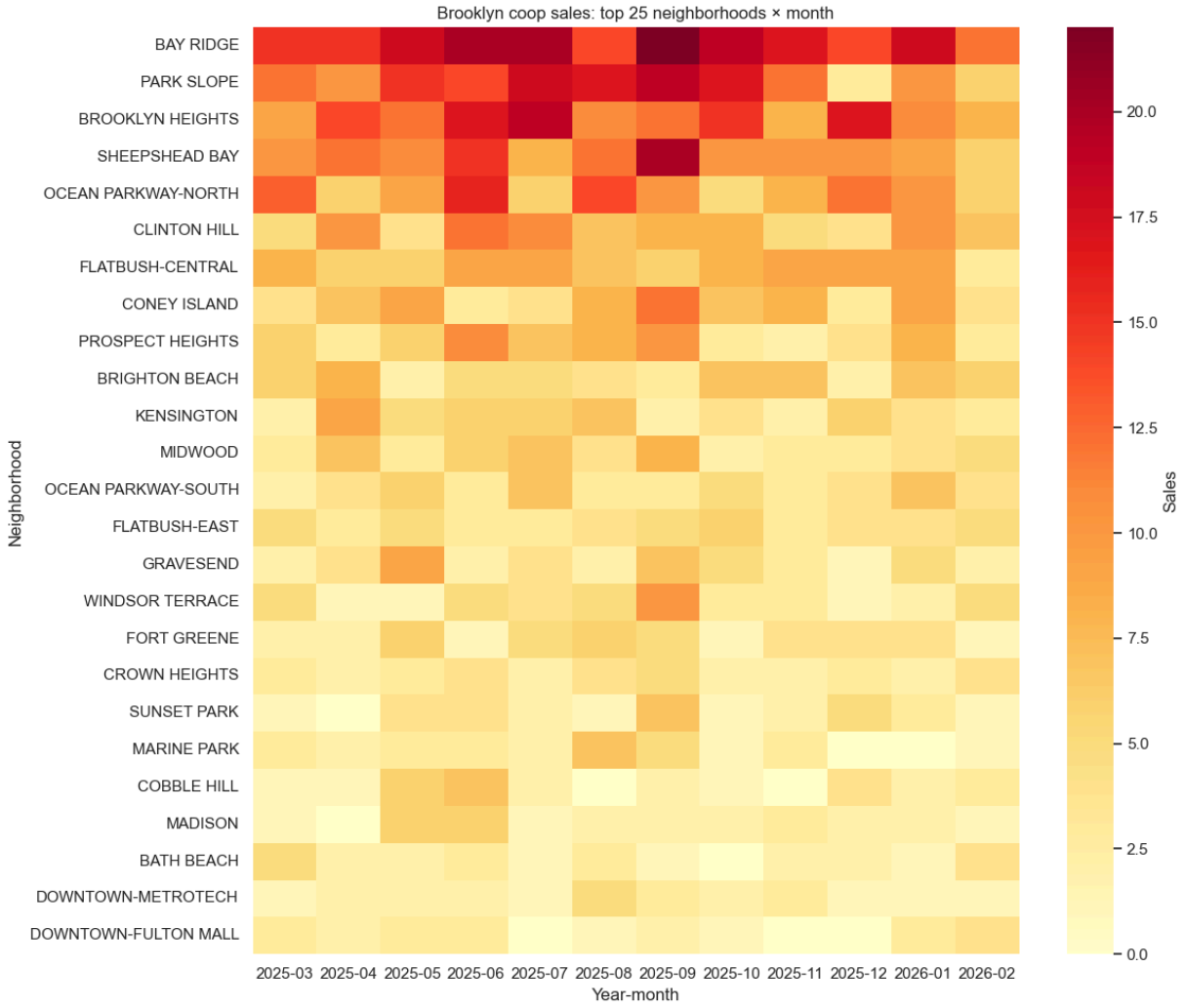
```
    print(
        "Calendar month (pooled across years): count and median price — high
    )
    print(by_cal_month.sort_values("median_price", ascending=False).to_strir

    MIN_N_PERIOD = max(10, int(len(coop_tp) * 0.01))
    by_period = coop_tp.groupby("YEAR_MONTH", observed=True).agg(
        n=("SALE PRICE", "count"),
        median_price=("SALE PRICE", "median"),
        mean_price=("SALE PRICE", "mean"),
    )
    top_periods = (
        by_period[by_period["n"] >= MIN_N_PERIOD].sort_values("median_price"
    )
    print(f"\nTop year-months by median price (min {MIN_N_PERIOD} sales):\n"
    print(top_periods.to_string())

    MIN_N_CELL = 5
    nb_ym = (
        coop_tp.groupby(["NEIGHBORHOOD", "YEAR_MONTH"], observed=True)["SALE
        .agg(n="count", median="median")
        .reset_index()
    )
    hot_cells = nb_ym[nb_ym["n"] >= MIN_N_CELL].sort_values("median", ascenc
    print(f"\nHighest median price by neighborhood × month (min {MIN_N_CELL}
    print(hot_cells.to_string(index=False))

    by_q = coop_tp.groupby("QUARTER", observed=True).agg(
        n=("SALE PRICE", "count"),
        median_price=("SALE PRICE", "median"),
    )
    print("\nBy quarter (top 12 by median):")
    print(by_q.sort_values("median_price", ascending=False).head(12).to_stri

    top15 = n_by_nbhd.head(15).index
    sub = coop_tp.loc[coop_tp["NEIGHBORHOOD"].isin(top15)]

    type_loc = (
        sub.groupby(["NEIGHBORHOOD", "COOP_TYPE"], observed=True)["SALE PRIC
        .agg(n="count", median="median")
        .unstack("COOP_TYPE")
    )
    print(f"\nTop 15 neighborhoods: sale counts and medians by coop type (cc
    print(type_loc.to_string())

    age_by_type = coop_tp.dropna(subset=["BUILDING_AGE"]).groupby("COOP_TYPE
    print("\nBuilding age (years) by coop type:")
    print(age_by_type.describe().to_string())

    fig, axes = plt.subplots(1, 3, figsize=(14, 4))

    coop_tp.loc[coop_tp["COOP_TYPE"] != "Other"].dropna(subset=["BUILDING_AG
        column="BUILDING_AGE", by="COOP_TYPE", ax=axes[0]
    )
    axes[0].set_title("Building age by coop type")
    axes[0].set_xlabel("Coop type")
```

```
    plt.suptitle("")

    for ax, ct in zip(axes[1:], ["Walkup", "Elevator"]):
        g = coop_tp.loc[coop_tp["COOP_TYPE"] == ct]
        g["NEIGHBORHOOD"].value_counts().head(12).sort_values().plot(kind="b
        ax.set_title(f"Top neighborhoods — {ct}")
        ax.set_xlabel("Sales")

    plt.tight_layout()
    plt.show()


for b in OUTER_BOROUGHS:
    analyze_borough_coops(b)
```

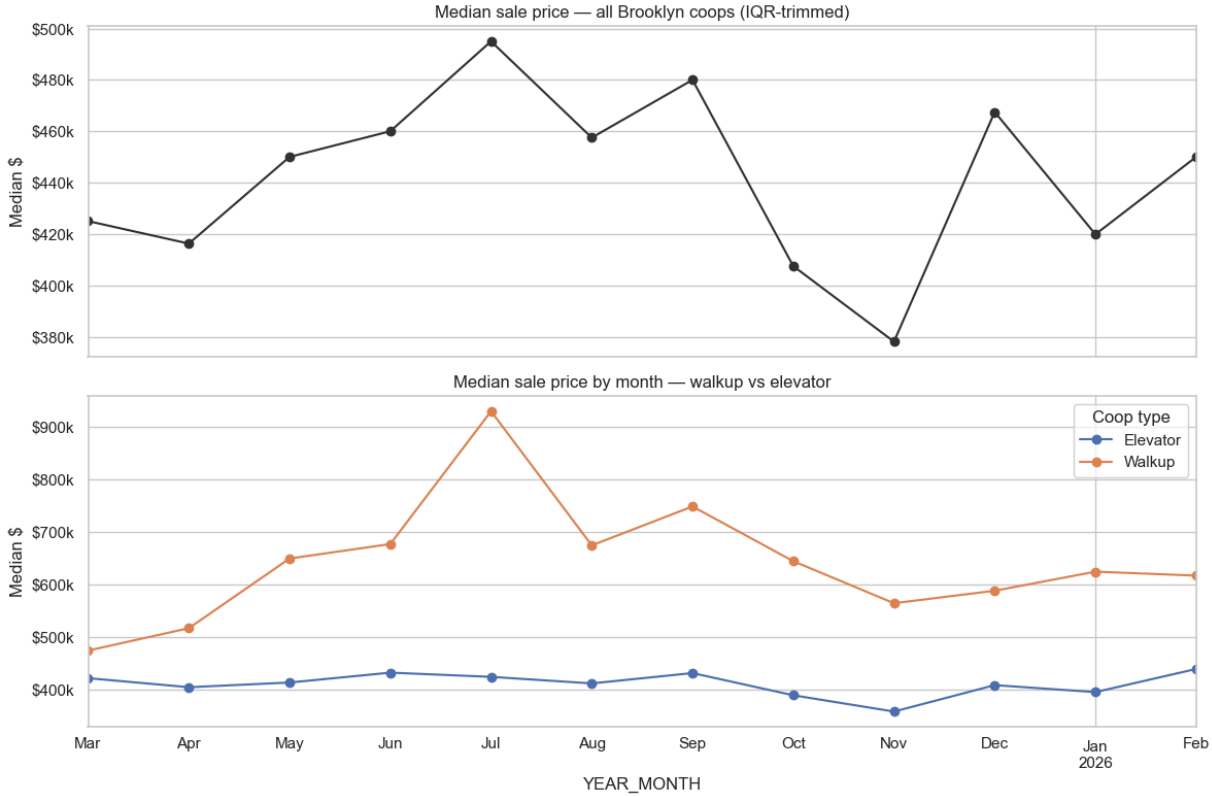
```
==========================================================================
 BROOKLYN COOPS
==========================================================================
Total coop sales (excludes CONDO COOPS and similar hybrids):
2233

Building class categories:
<StringArray>
['09 COOPS - WALKUP APARTMENTS', '10 COOPS - ELEVATOR APARTMENTS']
Length: 2, dtype: str

Columns:
Index(['NEIGHBORHOOD', 'BUILDING CLASS CATEGORY', 'TAX CLASS AT PRESENT',
       'BLOCK', 'LOT', 'BUILDING CLASS AT PRESENT', 'ADDRESS', 'ZIP CODE',
       'RESIDENTIAL UNITS', 'COMMERCIAL UNITS', 'TOTAL UNITS',
       'LAND SQUARE FEET', 'GROSS SQUARE FEET', 'YEAR BUILT',
       'TAX CLASS AT TIME OF SALE', 'BUILDING CLASS AT TIME OF SALE',
       'SALE PRICE', 'SALE DATE'],
      dtype='str')

Head:
    NEIGHBORHOOD       BUILDING CLASS CATEGORY TAX CLASS AT PRESENT  BLOCK
\
211   BATH BEACH  09 COOPS - WALKUP APARTMENTS                   2   6379
212   BATH BEACH  09 COOPS - WALKUP APARTMENTS                  2C   6459
213   BATH BEACH  09 COOPS - WALKUP APARTMENTS                   2   6462
214   BATH BEACH  09 COOPS - WALKUP APARTMENTS                   2   6462
215   BATH BEACH  09 COOPS - WALKUP APARTMENTS                   2   6462

     LOT BUILDING CLASS AT PRESENT                  ADDRESS  ZIP CODE  \
211   65                       C6  54 BAY 29TH STREET, A 12   11214.0
212    6                       C6      323 BAY 8 STREET, 2B   11228.0
213   40                       C6    1259 SHORE PARKWAY, C1   11214.0
214   40                       C6      1259 SHORE PKWY, A-3   11214.0
215   40                       C6  8847 BAY 16TH STREET, 2D   11214.0

     RESIDENTIAL UNITS  COMMERCIAL UNITS  TOTAL UNITS  LAND SQUARE FEET  \
211                NaN               NaN          NaN               NaN
212                NaN               NaN          NaN               NaN
213                NaN               NaN          NaN               NaN
214                NaN               NaN          NaN               NaN
215                NaN               NaN          NaN               NaN

     GROSS SQUARE FEET  YEAR BUILT  TAX CLASS AT TIME OF SALE  \
211                NaN      1926.0                           2
212                NaN      1950.0                           2
213                NaN      1949.0                           2
214                NaN      1949.0                           2
215                NaN      1949.0                           2

    BUILDING CLASS AT TIME OF SALE  SALE PRICE  SALE DATE
211                             C6      345000 2026-02-13
212                             C6      355000 2025-05-28
213                             C6      290000 2025-05-19
214                             C6      320000 2025-08-26
215                             C6      350000 2025-03-11
```

```
Entirely null columns (dropped): ['RESIDENTIAL UNITS', 'COMMERCIAL UNITS',
'TOTAL UNITS', 'LAND SQUARE FEET', 'GROSS SQUARE FEET']
    NEIGHBORHOOD       BUILDING CLASS CATEGORY TAX CLASS AT PRESENT  BLOCK
\
211   BATH BEACH  09 COOPS - WALKUP APARTMENTS                   2   6379
212   BATH BEACH  09 COOPS - WALKUP APARTMENTS                  2C   6459
213   BATH BEACH  09 COOPS - WALKUP APARTMENTS                   2   6462
214   BATH BEACH  09 COOPS - WALKUP APARTMENTS                   2   6462
215   BATH BEACH  09 COOPS - WALKUP APARTMENTS                   2   6462

     LOT BUILDING CLASS AT PRESENT                  ADDRESS  ZIP CODE  \
211   65                       C6  54 BAY 29TH STREET, A 12   11214.0
212    6                       C6      323 BAY 8 STREET, 2B   11228.0
213   40                       C6    1259 SHORE PARKWAY, C1   11214.0
214   40                       C6      1259 SHORE PKWY, A-3   11214.0
215   40                       C6  8847 BAY 16TH STREET, 2D   11214.0

     YEAR BUILT  TAX CLASS AT TIME OF SALE BUILDING CLASS AT TIME OF SALE  \
211      1926.0                         2                             C6
212      1950.0                         2                             C6
213      1949.0                         2                             C6
214      1949.0                         2                             C6
215      1949.0                         2                             C6

     SALE PRICE  SALE DATE
211      345000 2026-02-13
212      355000 2025-05-28
213      290000 2025-05-19
214      320000 2025-08-26
215      350000 2025-03-11

Fraction null per column:
NEIGHBORHOOD: 0.0
BUILDING CLASS CATEGORY: 0.0
TAX CLASS AT PRESENT: 0.0
BLOCK: 0.0
LOT: 0.0
BUILDING CLASS AT PRESENT: 0.0
ADDRESS: 0.0
ZIP CODE: 0.0
YEAR BUILT: 0.0
TAX CLASS AT TIME OF SALE: 0.0
BUILDING CLASS AT TIME OF SALE: 0.0
SALE PRICE: 0.0
SALE DATE: 0.0

After feature tweaks:
    NEIGHBORHOOD TAX CLASS AT PRESENT  BLOCK  LOT BUILDING CLASS AT PRESENT
\
211   BATH BEACH                   2   6379   65                       C6
212   BATH BEACH                  2C   6459    6                       C6
213   BATH BEACH                   2   6462   40                       C6
214   BATH BEACH                   2   6462   40                       C6
215   BATH BEACH                   2   6462   40                       C6
```

```
     ZIP CODE  YEAR BUILT BUILDING CLASS AT TIME OF SALE  SALE PRICE  \
211   11214.0      1926.0                             C6      345000
212   11228.0      1950.0                             C6      355000
213   11214.0      1949.0                             C6      290000
214   11214.0      1949.0                             C6      320000
215   11214.0      1949.0                             C6      350000

     SALE DATE  AGE OF THE BUILDING
211 2026-02-13               100.0
212 2025-05-28                76.0
213 2025-05-19                77.0
214 2025-08-26                77.0
215 2025-03-11                77.0

Fraction above upper IQR fence: 0.1495745633676668
    NEIGHBORHOOD TAX CLASS AT PRESENT  BLOCK  LOT BUILDING CLASS AT PRESENT
\
211   BATH BEACH                    2   6379   65                        C6
212   BATH BEACH                   2C   6459    6                        C6
213   BATH BEACH                    2   6462   40                        C6
214   BATH BEACH                    2   6462   40                        C6
215   BATH BEACH                    2   6462   40                        C6

     ZIP CODE  YEAR BUILT BUILDING CLASS AT TIME OF SALE  SALE PRICE  \
211   11214.0      1926.0                             C6      345000
212   11228.0      1950.0                             C6      355000
213   11214.0      1949.0                             C6      290000
214   11214.0      1949.0                             C6      320000
215   11214.0      1949.0                             C6      350000

     SALE DATE  AGE OF THE BUILDING
211 2026-02-13               100.0
212 2025-05-28                76.0
213 2025-05-19                77.0
214 2025-08-26                77.0
215 2025-03-11                77.0

COOP_TYPE counts:
COOP_TYPE
Elevator    1498
Walkup       401
Name: count, dtype: int64

Rows after upper IQR trim: 1,899
Sale date range: 2025-03-03 — 2026-02-27
Neighborhoods: 48
```


Brooklyn coop sales: top 25 neighborhoods × month
BAY RIDGE
PARK SLOPE
BROOKLYN HEIGHTS
SHEEPSHEAD BAY
OCEAN PARKWAY-NORTH
CLINTON HILL
FLATBUSH-CENTRAL
CONEY ISLAND
PROSPECT HEIGHTS
BRIGHTON BEACH
KENSINGTON
MIDWOOD
OCEAN PARKWAY-SOUTH
FLATBUSH-EAST
GRAVESEND
WINDSOR TERRACE
FORT GREENE
CROWN HEIGHTS
SUNSET PARK
MARINE PARK
COBBLE HILL
MADISON
BATH BEACH
DOWNTOWN-METROTECH
DOWNTOWN-FULTON MALL
Neighborhood
2025-03
2025-04
2025-05
2025-06
2025-07
2025-08
2025-09
2025-10
2025-11
2025-12
2026-01
2026-02
Year-month
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
Sales

```
Peak year-month per neighborhood (top 25 by volume) and share of that hood's
sales in that month:
                     peak_year_month  share_of_nbhd_in_peak_month
NEIGHBORHOOD
COBBLE HILL                  2025-06                     0.241379
MARINE PARK                  2025-08                     0.233333
SUNSET PARK                  2025-09                     0.225806
WINDSOR TERRACE              2025-09                     0.222222
MADISON                      2025-05                     0.214286
DOWNTOWN-METROTECH           2025-08                     0.208333
GRAVESEND                    2025-05                     0.195652
BATH BEACH                   2025-03                     0.192308
DOWNTOWN-FULTON MALL         2026-02                     0.181818
KENSINGTON                   2025-04                     0.160714
PROSPECT HEIGHTS             2025-06                     0.154930
CONEY ISLAND                 2025-09                     0.153846
SHEEPSHEAD BAY               2025-09                     0.150376
FORT GREENE                  2025-05                     0.146341
MIDWOOD                      2025-09                     0.145455
OCEAN PARKWAY-NORTH          2025-06                     0.139130
CROWN HEIGHTS                2025-09                     0.138889
OCEAN PARKWAY-SOUTH          2025-07                     0.137255
CLINTON HILL                 2025-06                     0.131868
BRIGHTON BEACH               2025-04                     0.129032
PARK SLOPE                   2025-09                     0.124183
BROOKLYN HEIGHTS             2025-07                     0.124183
FLATBUSH-EAST                2025-10                     0.120000
BAY RIDGE                    2025-09                     0.107843
FLATBUSH-CENTRAL             2025-12                     0.101124
```



Calendar month (pooled across years): count and median price — higher median ≈ pricier time of year in this sample

month_name	n	median_price
July	176	495000.0
September	201	480000.0
December	130	467500.0
June	193	460000.0
August	169	457500.0
February	121	450000.0
May	166	450000.0
March	143	425000.0
January	164	419857.5
April	146	416250.0
October	154	407500.0
November	136	378062.0

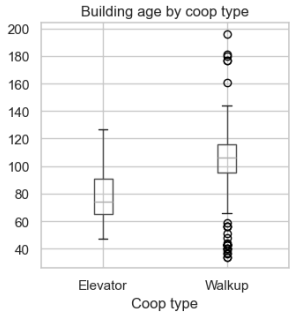
Top year-months by median price (min 18 sales):

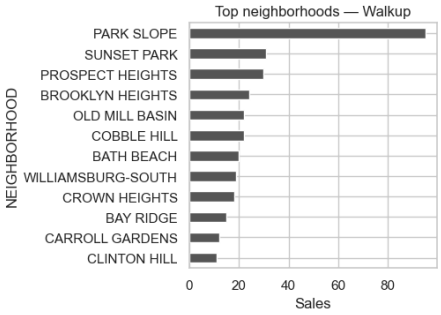
YEAR_MONTH	n	median_price	mean_price
2025-07	176	495000.0	643330.250000
2025-09	201	480000.0	592124.542289
2025-12	130	467500.0	541864.092308
2025-06	193	460000.0	595672.761658
2025-08	169	457500.0	574617.112426
2025-05	166	450000.0	546561.078313
2026-02	121	450000.0	574131.818182
2025-03	143	425000.0	523825.223776
2026-01	164	419857.5	542824.719512
2025-04	146	416250.0	535628.493151
2025-10	154	407500.0	523648.694805
2025-11	136	378062.0	475935.602941

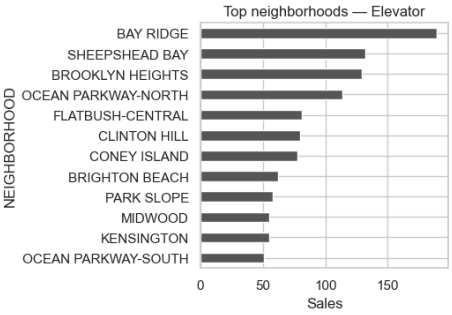
Highest median price by neighborhood × month (min 5 sales):

NEIGHBORHOOD	YEAR_MONTH	n	median
PARK SLOPE	2025-07	18	1150000.0
PARK SLOPE	2026-02	6	1057500.0
CLINTON HILL	2026-02	7	1050000.0
PROSPECT HEIGHTS	2025-08	8	1047500.0
PARK SLOPE	2026-01	10	1045000.0
PROSPECT HEIGHTS	2025-07	7	1015000.0
WINDSOR TERRACE	2026-02	5	999000.0
PARK SLOPE	2025-08	17	981000.0
PARK SLOPE	2025-05	15	970000.0
BROOKLYN HEIGHTS	2025-08	11	950000.0
PROSPECT HEIGHTS	2025-06	11	940000.0
PARK SLOPE	2025-09	19	938000.0
CLINTON HILL	2025-09	8	934750.0
FORT GREENE	2025-09	5	929000.0
BENSONHURST	2026-01	6	901630.0
PARK SLOPE	2025-03	12	892500.0
PROSPECT HEIGHTS	2025-05	6	888000.0
PROSPECT HEIGHTS	2025-09	10	884500.0
PARK SLOPE	2025-11	12	882500.0
BROOKLYN HEIGHTS	2026-01	11	850000.0

```
BROOKLYN HEIGHTS    2025-07 19  850000.0
PROSPECT HEIGHTS    2026-01  8  847500.0
      PARK SLOPE    2025-04 10  842000.0
 WINDSOR TERRACE    2025-09 10  837250.0
    CLINTON HILL    2025-10  8  830000.0

By quarter (top 12 by median):
           n  median_price
QUARTER
2025Q3   546      475000.0
2025Q2   505      448000.0
2026Q1   285      440000.0
2025Q1   143      425000.0
2025Q4   420      415000.0

Top 15 neighborhoods: sale counts and medians by coop type (columns = type)
— Brooklyn
                          n           median
COOP_TYPE           Elevator Walkup  Elevator     Walkup
NEIGHBORHOOD
BAY RIDGE              189.0   15.0  350000.0   375000.0
BRIGHTON BEACH          62.0    NaN  325000.0        NaN
BROOKLYN HEIGHTS       129.0   24.0  700000.0  1022500.0
CLINTON HILL            80.0   11.0  805000.0   970000.0
CONEY ISLAND            78.0    NaN  370000.0        NaN
FLATBUSH-CENTRAL        81.0    8.0  495000.0   575000.0
FLATBUSH-EAST           50.0    NaN  257000.0        NaN
GRAVESEND               45.0    1.0  365000.0   300000.0
KENSINGTON              55.0    1.0  541500.0   400000.0
MIDWOOD                 55.0    NaN  330000.0        NaN
OCEAN PARKWAY-NORTH    114.0    1.0  401000.0   150000.0
OCEAN PARKWAY-SOUTH     51.0    NaN  280000.0        NaN
PARK SLOPE              58.0   95.0  782500.0   981000.0
PROSPECT HEIGHTS        41.0   30.0  940000.0   865500.0
SHEEPSHEAD BAY         132.0    1.0  282100.0   219000.0

Building age (years) by coop type:
            count        mean        std   min   25%    50%    75%    max
COOP_TYPE
Elevator   1498.0   78.361148  15.117832  47.0  65.0   74.0   91.0  127.0
Walkup      401.0  102.708229  22.428044  34.0  95.0  106.0  116.0  196.0
```



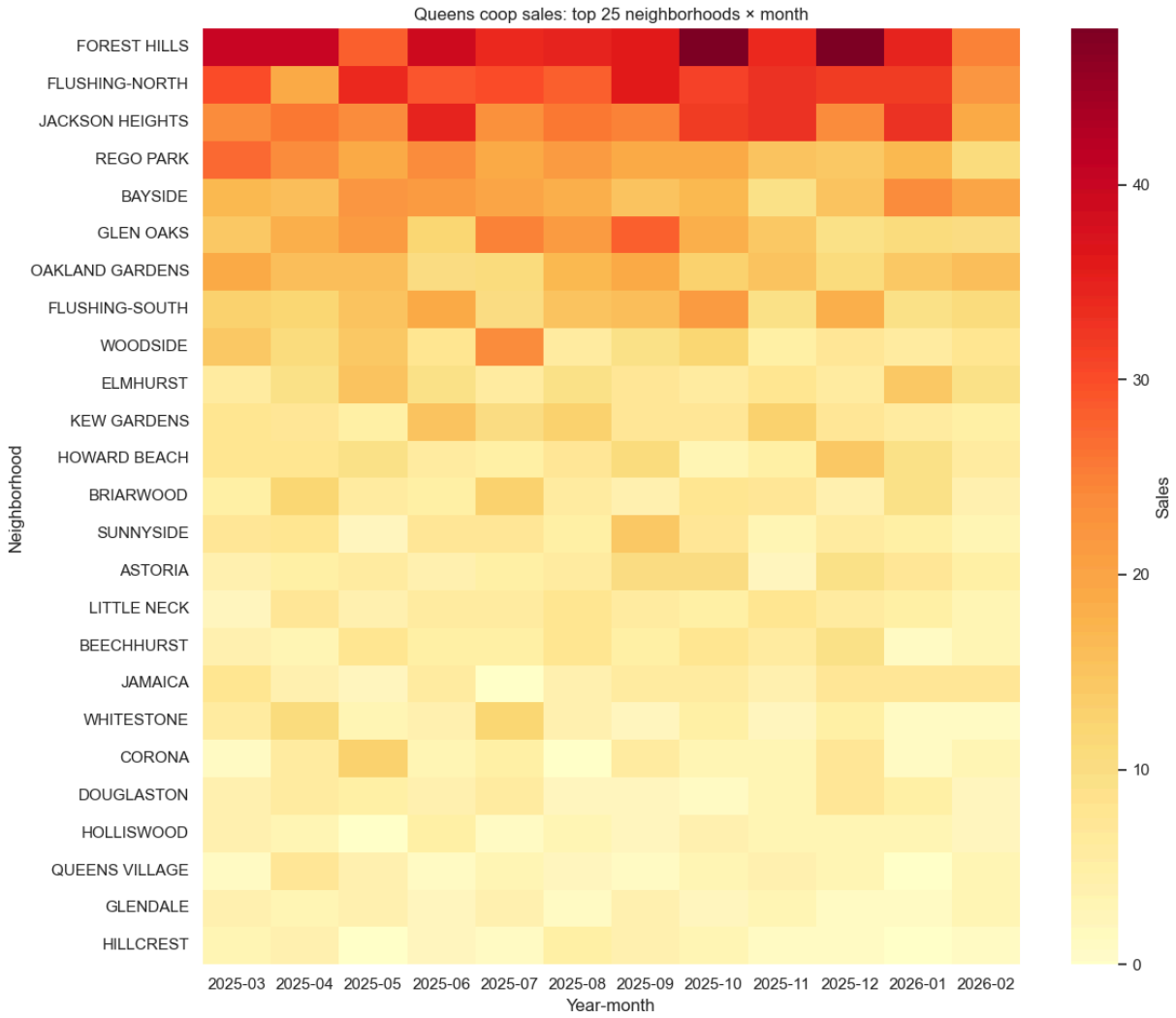




```
==========================================================================
 QUEENS COOPS
==========================================================================
Total coop sales (excludes CONDO COOPS and similar hybrids):
3724

Building class categories:
<StringArray>
['09 COOPS - WALKUP APARTMENTS', '10 COOPS - ELEVATOR APARTMENTS']
Length: 2, dtype: str

Columns:
Index(['NEIGHBORHOOD', 'BUILDING CLASS CATEGORY', 'TAX CLASS AT PRESENT',
       'BLOCK', 'LOT', 'BUILDING CLASS AT PRESENT', 'ADDRESS', 'ZIP CODE',
       'RESIDENTIAL UNITS', 'COMMERCIAL UNITS', 'TOTAL UNITS',
       'LAND SQUARE FEET', 'GROSS SQUARE FEET', 'YEAR BUILT',
       'TAX CLASS AT TIME OF SALE', 'BUILDING CLASS AT TIME OF SALE',
       'SALE PRICE', 'SALE DATE'],
      dtype='str')

Head:
    NEIGHBORHOOD       BUILDING CLASS CATEGORY TAX CLASS AT PRESENT  BLOCK
\
927      ASTORIA  09 COOPS - WALKUP APARTMENTS                   2    569
929      ASTORIA  09 COOPS - WALKUP APARTMENTS                   2    578
930      ASTORIA  09 COOPS - WALKUP APARTMENTS                   2    578
931      ASTORIA  09 COOPS - WALKUP APARTMENTS                   2    578
933      ASTORIA  09 COOPS - WALKUP APARTMENTS                   2    654

     LOT BUILDING CLASS AT PRESENT                    ADDRESS  ZIP CODE  \
927   56                       C6         31-18 CRESCENT, 5A   11106.0
929   20                       C6  30-83 CRESCENT STREET, A3   11102.0
930   20                       C6  30-83 CRESCENT STREET, B1   11102.0
931   20                       C6  30-83 CRESCENT STREET, B2   11102.0
933   24                       C6       24-75 38 STREET, A10   11103.0

     RESIDENTIAL UNITS  COMMERCIAL UNITS  TOTAL UNITS  LAND SQUARE FEET  \
927                NaN               NaN          NaN               NaN
929                NaN               NaN          NaN               NaN
930                NaN               NaN          NaN               NaN
931                NaN               NaN          NaN               NaN
933                NaN               NaN          NaN               NaN

     GROSS SQUARE FEET  YEAR BUILT  TAX CLASS AT TIME OF SALE  \
927                NaN      1926.0                           2
929                NaN      1928.0                           2
930                NaN      1928.0                           2
931                NaN      1928.0                           2
933                NaN      1929.0                           2

    BUILDING CLASS AT TIME OF SALE  SALE PRICE  SALE DATE
927                             C6      200000 2025-05-15
929                             C6      360000 2025-10-16
930                             C6      520000 2025-08-12
931                             C6      345000 2026-02-18
933                             C6      738000 2026-02-24
```

```
Entirely null columns (dropped): ['RESIDENTIAL UNITS', 'COMMERCIAL UNITS',
'TOTAL UNITS', 'LAND SQUARE FEET', 'GROSS SQUARE FEET']
    NEIGHBORHOOD       BUILDING CLASS CATEGORY TAX CLASS AT PRESENT  BLOCK
\
927      ASTORIA  09 COOPS - WALKUP APARTMENTS                   2    569
929      ASTORIA  09 COOPS - WALKUP APARTMENTS                   2    578
930      ASTORIA  09 COOPS - WALKUP APARTMENTS                   2    578
931      ASTORIA  09 COOPS - WALKUP APARTMENTS                   2    578
933      ASTORIA  09 COOPS - WALKUP APARTMENTS                   2    654

     LOT BUILDING CLASS AT PRESENT                   ADDRESS  ZIP CODE  \
927   56                       C6         31-18 CRESCENT, 5A   11106.0
929   20                       C6  30-83 CRESCENT STREET, A3   11102.0
930   20                       C6  30-83 CRESCENT STREET, B1   11102.0
931   20                       C6  30-83 CRESCENT STREET, B2   11102.0
933   24                       C6       24-75 38 STREET, A10   11103.0

     YEAR BUILT  TAX CLASS AT TIME OF SALE BUILDING CLASS AT TIME OF SALE  \
927      1926.0                         2                              C6
929      1928.0                         2                              C6
930      1928.0                         2                              C6
931      1928.0                         2                              C6
933      1929.0                         2                              C6

     SALE PRICE  SALE DATE
927      200000 2025-05-15
929      360000 2025-10-16
930      520000 2025-08-12
931      345000 2026-02-18
933      738000 2026-02-24

Fraction null per column:
NEIGHBORHOOD: 0.0
BUILDING CLASS CATEGORY: 0.0
TAX CLASS AT PRESENT: 0.0
BLOCK: 0.0
LOT: 0.0
BUILDING CLASS AT PRESENT: 0.0
ADDRESS: 0.0
ZIP CODE: 0.002953813104189044
YEAR BUILT: 0.0002685284640171858
TAX CLASS AT TIME OF SALE: 0.0
BUILDING CLASS AT TIME OF SALE: 0.0
SALE PRICE: 0.0
SALE DATE: 0.0

After feature tweaks:
    NEIGHBORHOOD TAX CLASS AT PRESENT  BLOCK  LOT BUILDING CLASS AT PRESENT
\
927      ASTORIA                    2    569   56                       C6
929      ASTORIA                    2    578   20                       C6
930      ASTORIA                    2    578   20                       C6
931      ASTORIA                    2    578   20                       C6
933      ASTORIA                    2    654   24                       C6
```

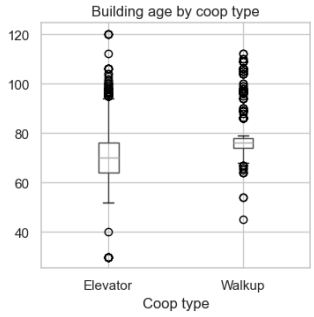
```
     ZIP CODE  YEAR BUILT BUILDING CLASS AT TIME OF SALE  SALE PRICE  \
927   11106.0      1926.0                             C6      200000
929   11102.0      1928.0                             C6      360000
930   11102.0      1928.0                             C6      520000
931   11102.0      1928.0                             C6      345000
933   11103.0      1929.0                             C6      738000

     SALE DATE  AGE OF THE BUILDING
927 2025-05-15               100.0
929 2025-10-16                98.0
930 2025-08-12                98.0
931 2026-02-18                98.0
933 2026-02-24                97.0

Fraction above upper IQR fence: 0.0985499462943072
    NEIGHBORHOOD TAX CLASS AT PRESENT  BLOCK  LOT BUILDING CLASS AT PRESENT
\
927      ASTORIA                    2    569   56                        C6
929      ASTORIA                    2    578   20                        C6
930      ASTORIA                    2    578   20                        C6
931      ASTORIA                    2    578   20                        C6
934      ASTORIA                    2    654   24                        C6

     ZIP CODE  YEAR BUILT BUILDING CLASS AT TIME OF SALE  SALE PRICE  \
927   11106.0      1926.0                             C6      200000
929   11102.0      1928.0                             C6      360000
930   11102.0      1928.0                             C6      520000
931   11102.0      1928.0                             C6      345000
934   11103.0      1929.0                             C6      370000

     SALE DATE  AGE OF THE BUILDING
927 2025-05-15               100.0
929 2025-10-16                98.0
930 2025-08-12                98.0
931 2026-02-18                98.0
934 2025-07-16                97.0

COOP_TYPE counts:
COOP_TYPE
Elevator    2525
Walkup       832
Name: count, dtype: int64

Rows after upper IQR trim: 3,357
Sale date range: 2025-03-03 — 2026-02-27
Neighborhoods: 43
```

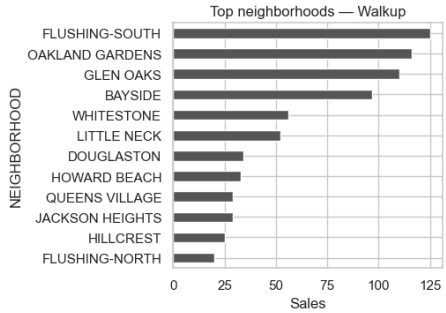

Queens coop sales: top 25 neighborhoods × month
FOREST HILLS
FLUSHING-NORTH
JACKSON HEIGHTS
REGO PARK
BAYSIDE
GLEN OAKS
OAKLAND GARDENS
FLUSHING-SOUTH
WOODSIDE
ELMHURST
KEW GARDENS
HOWARD BEACH
BRIARWOOD
SUNNYSIDE
ASTORIA
LITTLE NECK
BEECHHURST
JAMAICA
WHITESTONE
CORONA
DOUGLASTON
HOLLISWOOD
QUEENS VILLAGE
GLENDALE
HILLCREST
Neighborhood
2025-03
2025-04
2025-05
2025-06
2025-07
2025-08
2025-09
2025-10
2025-11
2025-12
2026-01
2026-02
Year-month
Sales
0
10
20
30
40

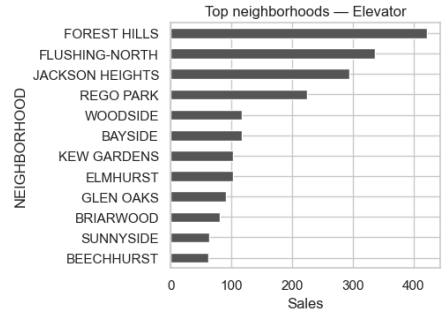
```
Peak year-month per neighborhood (top 25 by volume) and share of that hood's
sales in that month:
                peak_year_month  share_of_nbhd_in_peak_month
NEIGHBORHOOD
CORONA                  2025-05                     0.254902
QUEENS VILLAGE          2025-04                     0.218750
WHITESTONE              2025-07                     0.214286
HILLCREST               2025-08                     0.200000
WOODSIDE                2025-07                     0.193548
SUNNYSIDE               2025-09                     0.189189
BRIARWOOD               2025-07                     0.156627
HOWARD BEACH            2025-12                     0.153846
HOLLISWOOD              2025-06                     0.151515
DOUGLASTON              2025-12                     0.148936
KEW GARDENS             2025-06                     0.145631
ELMHURST                2025-05                     0.144231
GLEN OAKS               2025-09                     0.139303
BEECHHURST              2025-12                     0.138462
ASTORIA                 2025-10                     0.136986
JAMAICA                 2025-03                     0.131148
GLENDALE                2025-07                     0.125000
FLUSHING-SOUTH          2025-10                     0.125000
LITTLE NECK             2025-08                     0.121212
REGO PARK               2025-03                     0.117904
BAYSIDE                 2026-01                     0.112150
FOREST HILLS            2025-12                     0.108597
JACKSON HEIGHTS         2025-06                     0.108025
OAKLAND GARDENS         2025-03                     0.107345
FLUSHING-NORTH          2025-09                     0.101124
```

```
Calendar month (pooled across years): count and median price — higher median
≈ pricier time of year in this sample
              n  median_price
month_name
February    206      325500.0
June        293      325000.0
July        301      325000.0
August      282      325000.0
October     306      325000.0
November    249      325000.0
May         284      323500.0
January     264      322500.0
December    286      320500.0
March       280      320000.0
April       298      320000.0
September   308      315000.0

Top year-months by median price (min 33 sales):

              n  median_price     mean_price
YEAR_MONTH
2026-02     206      325500.0  336607.747573
2025-06     293      325000.0  338994.290102
2025-07     301      325000.0  328439.491694
2025-08     282      325000.0  340468.177305
2025-10     306      325000.0  344263.558824
2025-11     249      325000.0  337627.325301
2025-05     284      323500.0  339705.281690
2026-01     264      322500.0  334511.890152
2025-12     286      320500.0  326530.146853
2025-03     280      320000.0  330567.682143
2025-04     298      320000.0  330781.875839
2025-09     308      315000.0  335650.090909

Highest median price by neighborhood × month (min 5 sales):
NEIGHBORHOOD YEAR_MONTH  n   median
     ASTORIA    2026-01  7 556000.0
     ASTORIA    2025-12  9 526435.0
     ASTORIA    2025-10 10 522500.0
     ASTORIA    2025-08  6 502500.0
     ASTORIA    2025-09 10 488360.0
      CORONA    2025-05 13 487605.0
     ASTORIA    2025-04  5 460000.0
     ASTORIA    2026-02  5 460000.0
  BEECHHURST    2025-09  5 449000.0
  BEECHHURST    2025-10  8 435000.0
   GLEN OAKS    2025-12  9 425900.0
   SUNNYSIDE    2025-10  7 425000.0
  BEECHHURST    2025-07  5 425000.0
     ASTORIA    2025-05  6 420000.0
  BEECHHURST    2025-08  8 419500.0
  BEECHHURST    2025-05  8 417500.0
   SUNNYSIDE    2025-03  7 415000.0
   SUNNYSIDE    2025-08  5 412500.0
   GLEN OAKS    2025-06 12 410000.0
     BAYSIDE    2025-11  9 410000.0
```

```
   SUNNYSIDE    2025-09 14 405000.0
 LITTLE NECK    2025-06  6 402500.0
   GLEN OAKS    2025-03 14 400000.0
   GLEN OAKS    2025-07 25 400000.0
    WOODSIDE    2025-03 14 399500.0

By quarter (top 12 by median):
           n  median_price
QUARTER
2025Q2   875      325000.0
2025Q4   841      325000.0
2026Q1   470      325000.0
2025Q1   280      320000.0
2025Q3   891      320000.0

Top 15 neighborhoods: sale counts and medians by coop type (columns = type)
— Queens
                       n           median
COOP_TYPE       Elevator Walkup  Elevator    Walkup
NEIGHBORHOOD
ASTORIA             60.0   13.0  491000.0  415000.0
BAYSIDE            117.0   97.0  370000.0  305000.0
BRIARWOOD           81.0    2.0  270000.0  170000.0
ELMHURST           103.0    1.0  320000.0  351000.0
FLUSHING-NORTH     336.0   20.0  342900.0  255000.0
FLUSHING-SOUTH      43.0  125.0  270000.0  299000.0
FOREST HILLS       423.0   19.0  340000.0  290204.0
GLEN OAKS           91.0  110.0  425000.0  342000.0
HOWARD BEACH        58.0   33.0  222000.0  330000.0
JACKSON HEIGHTS    295.0   29.0  332500.0  415000.0
KEW GARDENS        103.0    NaN  275000.0       NaN
OAKLAND GARDENS     61.0  116.0  260000.0  348500.0
REGO PARK          225.0    4.0  320000.0  235000.0
SUNNYSIDE           64.0   10.0  405000.0  384999.5
WOODSIDE           117.0    7.0  330000.0  325000.0

Building age (years) by coop type:
            count       mean        std   min   25%   50%   75%    max
COOP_TYPE
Elevator   2525.0  71.666931  11.057136  30.0  64.0  70.0  76.0  120.0
Walkup      832.0  77.918269   8.501182  45.0  74.0  76.0  78.0  112.0
```



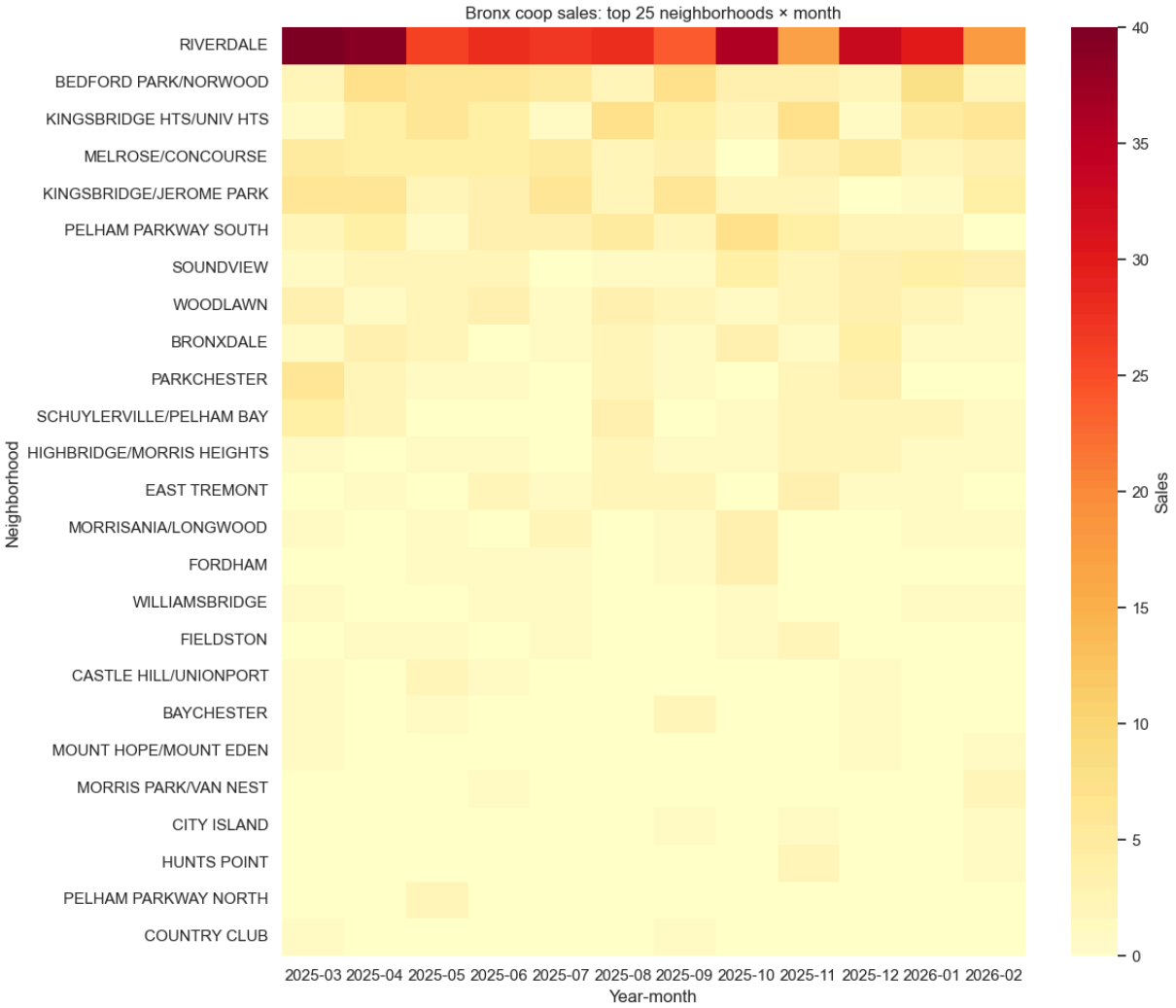




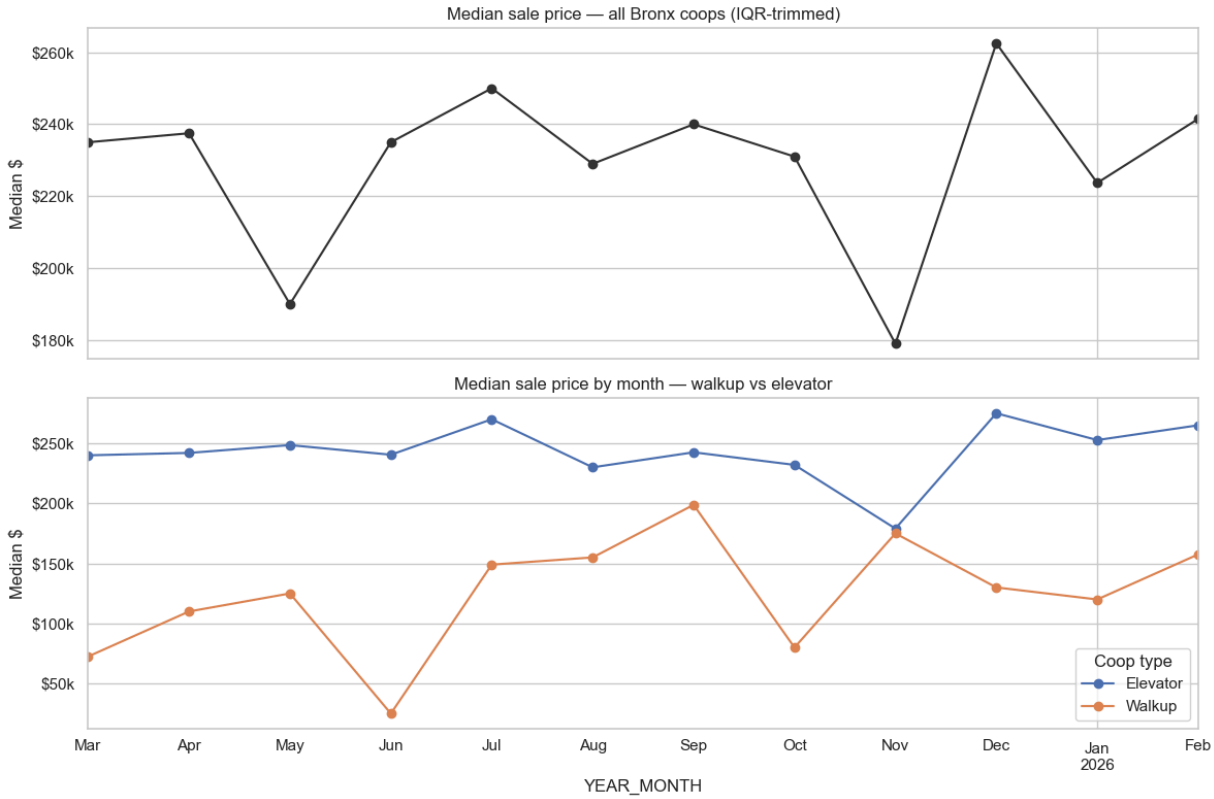
```
==========================================================================
 BRONX COOPS
==========================================================================
Total coop sales (excludes CONDO COOPS and similar hybrids):
817

Building class categories:
<StringArray>
['10 COOPS - ELEVATOR APARTMENTS', '09 COOPS - WALKUP APARTMENTS']
Length: 2, dtype: str

Columns:
Index(['NEIGHBORHOOD', 'BUILDING CLASS CATEGORY', 'TAX CLASS AT PRESENT',
       'BLOCK', 'LOT', 'BUILDING CLASS AT PRESENT', 'ADDRESS', 'ZIP CODE',
       'RESIDENTIAL UNITS', 'COMMERCIAL UNITS', 'TOTAL UNITS',
       'LAND SQUARE FEET', 'GROSS SQUARE FEET', 'YEAR BUILT',
       'TAX CLASS AT TIME OF SALE', 'BUILDING CLASS AT TIME OF SALE',
       'SALE PRICE', 'SALE DATE'],
      dtype='str')

Head:
    NEIGHBORHOOD         BUILDING CLASS CATEGORY TAX CLASS AT PRESENT  BLOCK
\
41      BATHGATE  10 COOPS - ELEVATOR APARTMENTS                   2   3044
42      BATHGATE  10 COOPS - ELEVATOR APARTMENTS                   2   3044
431   BAYCHESTER  10 COOPS - ELEVATOR APARTMENTS                   2   5259
432   BAYCHESTER  10 COOPS - ELEVATOR APARTMENTS                   2   5259
433   BAYCHESTER  10 COOPS - ELEVATOR APARTMENTS                   2   5259

     LOT BUILDING CLASS AT PRESENT                     ADDRESS  ZIP CODE  \
41    40                       D4          495 E 178TH ST, 3G     10457
42    40                       D4      495 E 178TH STREET, 3C     10457
431   16                       D4  3410 DE REIMER AVENUE, #6K     10475
432   16                       D4  3410 DE REIMER AVENUE, 11H     10475
433   16                       D4    3410 DEREIMER AVENUE, 9K     10475

     RESIDENTIAL UNITS  COMMERCIAL UNITS  TOTAL UNITS  LAND SQUARE FEET  \
41                 NaN               NaN          NaN               NaN
42                 NaN               NaN          NaN               NaN
431                NaN               NaN          NaN               NaN
432                NaN               NaN          NaN               NaN
433                NaN               NaN          NaN               NaN

     GROSS SQUARE FEET  YEAR BUILT  TAX CLASS AT TIME OF SALE  \
41                 NaN      1941.0                           2
42                 NaN      1941.0                           2
431                NaN      1965.0                           2
432                NaN      1965.0                           2
433                NaN      1965.0                           2

    BUILDING CLASS AT TIME OF SALE  SALE PRICE  SALE DATE
41                               D4      110000 2025-09-03
42                               D4       70000 2025-05-30
431                              D4      360000 2025-05-09
432                              D4      112000 2025-09-25
433                              D4      329000 2025-03-04
```

```
Entirely null columns (dropped): ['RESIDENTIAL UNITS', 'COMMERCIAL UNITS',
'TOTAL UNITS', 'LAND SQUARE FEET', 'GROSS SQUARE FEET']
    NEIGHBORHOOD         BUILDING CLASS CATEGORY TAX CLASS AT PRESENT  BLOCK
\
41      BATHGATE  10 COOPS - ELEVATOR APARTMENTS                   2   3044
42      BATHGATE  10 COOPS - ELEVATOR APARTMENTS                   2   3044
431   BAYCHESTER  10 COOPS - ELEVATOR APARTMENTS                   2   5259
432   BAYCHESTER  10 COOPS - ELEVATOR APARTMENTS                   2   5259
433   BAYCHESTER  10 COOPS - ELEVATOR APARTMENTS                   2   5259

     LOT BUILDING CLASS AT PRESENT                     ADDRESS  ZIP CODE  \
41    40                       D4          495 E 178TH ST, 3G     10457
42    40                       D4      495 E 178TH STREET, 3C     10457
431   16                       D4  3410 DE REIMER AVENUE, #6K     10475
432   16                       D4  3410 DE REIMER AVENUE, 11H     10475
433   16                       D4    3410 DEREIMER AVENUE, 9K     10475

     YEAR BUILT  TAX CLASS AT TIME OF SALE BUILDING CLASS AT TIME OF SALE  \
41       1941.0                         2                             D4
42       1941.0                         2                             D4
431      1965.0                         2                             D4
432      1965.0                         2                             D4
433      1965.0                         2                             D4

     SALE PRICE  SALE DATE
41       110000 2025-09-03
42        70000 2025-05-30
431      360000 2025-05-09
432      112000 2025-09-25
433      329000 2025-03-04

Fraction null per column:
NEIGHBORHOOD: 0.0
BUILDING CLASS CATEGORY: 0.0
TAX CLASS AT PRESENT: 0.0
BLOCK: 0.0
LOT: 0.0
BUILDING CLASS AT PRESENT: 0.0
ADDRESS: 0.0
ZIP CODE: 0.0
YEAR BUILT: 0.0
TAX CLASS AT TIME OF SALE: 0.0
BUILDING CLASS AT TIME OF SALE: 0.0
SALE PRICE: 0.0
SALE DATE: 0.0

After feature tweaks:
    NEIGHBORHOOD TAX CLASS AT PRESENT  BLOCK  LOT BUILDING CLASS AT PRESENT
\
41      BATHGATE                    2   3044   40                       D4
42      BATHGATE                    2   3044   40                       D4
431   BAYCHESTER                    2   5259   16                       D4
432   BAYCHESTER                    2   5259   16                       D4
433   BAYCHESTER                    2   5259   16                       D4
```

```
     ZIP CODE  YEAR BUILT BUILDING CLASS AT TIME OF SALE  SALE PRICE  \
41      10457      1941.0                             D4      110000
42      10457      1941.0                             D4       70000
431     10475      1965.0                             D4      360000
432     10475      1965.0                             D4      112000
433     10475      1965.0                             D4      329000

     SALE DATE  AGE OF THE BUILDING
41  2025-09-03                85.0
42  2025-05-30                85.0
431 2025-05-09                61.0
432 2025-09-25                61.0
433 2025-03-04                61.0

Fraction above upper IQR fence: 0.07833537331701346
    NEIGHBORHOOD TAX CLASS AT PRESENT  BLOCK  LOT BUILDING CLASS AT PRESENT
\
41      BATHGATE                    2   3044   40                        D4
42      BATHGATE                    2   3044   40                        D4
431   BAYCHESTER                    2   5259   16                        D4
432   BAYCHESTER                    2   5259   16                        D4
433   BAYCHESTER                    2   5259   16                        D4

     ZIP CODE  YEAR BUILT BUILDING CLASS AT TIME OF SALE  SALE PRICE  \
41      10457      1941.0                             D4      110000
42      10457      1941.0                             D4       70000
431     10475      1965.0                             D4      360000
432     10475      1965.0                             D4      112000
433     10475      1965.0                             D4      329000

     SALE DATE  AGE OF THE BUILDING
41  2025-09-03                85.0
42  2025-05-30                85.0
431 2025-05-09                61.0
432 2025-09-25                61.0
433 2025-03-04                61.0

COOP_TYPE counts:
COOP_TYPE
Elevator    687
Walkup       66
Name: count, dtype: int64

Rows after upper IQR trim: 753
Sale date range: 2025-03-03 — 2026-02-27
Neighborhoods: 28
```


Bronx coop sales: top 25 neighborhoods × month
RIVERDALE
BEDFORD PARK/NORWOOD
KINGSBRIDGE HTS/UNIV HTS
MELROSE/CONCOURSE
KINGSBRIDGE/JEROME PARK
PELHAM PARKWAY SOUTH
SOUNDVIEW
WOODLAWN
BRONXDALE
PARKCHESTER
SCHUYLERVILLE/PELHAM BAY
HIGHBRIDGE/MORRIS HEIGHTS
EAST TREMONT
MORRISANIA/LONGWOOD
FORDHAM
WILLIAMSBRIDGE
FIELDSTON
CASTLE HILL/UNIONPORT
BAYCHESTER
MOUNT HOPE/MOUNT EDEN
MORRIS PARK/VAN NEST
CITY ISLAND
HUNTS POINT
PELHAM PARKWAY NORTH
COUNTRY CLUB
Neighborhood
2025-03 2025-04 2025-05 2025-06 2025-07 2025-08 2025-09 2025-10 2025-11 2025-12 2026-01 2026-02
Year-month
40
35
30
25
20
15
10
5
0
Sales

```
Peak year-month per neighborhood (top 25 by volume) and share of that hood's
sales in that month:
                          peak_year_month  share_of_nbhd_in_peak_month
NEIGHBORHOOD
PELHAM PARKWAY NORTH              2025-05                     1.000000
HUNTS POINT                       2025-11                     0.666667
MORRIS PARK/VAN NEST              2026-02                     0.666667
COUNTRY CLUB                      2025-03                     0.500000
FORDHAM                           2025-10                     0.428571
BAYCHESTER                        2025-09                     0.400000
CASTLE HILL/UNIONPORT             2025-05                     0.400000
FIELDSTON                         2025-11                     0.333333
CITY ISLAND                       2025-09                     0.333333
MOUNT HOPE/MOUNT EDEN             2026-02                     0.333333
PARKCHESTER                       2025-03                     0.333333
MORRISANIA/LONGWOOD               2025-10                     0.300000
SCHUYLERVILLE/PELHAM BAY          2025-03                     0.235294
EAST TREMONT                      2025-11                     0.230769
BRONXDALE                         2025-12                     0.200000
PELHAM PARKWAY SOUTH              2025-10                     0.200000
WILLIAMSBRIDGE                    2025-10                     0.166667
SOUNDVIEW                         2026-01                     0.160000
HIGHBRIDGE/MORRIS HEIGHTS         2025-12                     0.153846
BEDFORD PARK/NORWOOD              2026-01                     0.150943
KINGSBRIDGE/JEROME PARK           2025-07                     0.150000
KINGSBRIDGE HTS/UNIV HTS          2025-11                     0.145833
MELROSE/CONCOURSE                 2025-07                     0.125000
WOODLAWN                          2025-08                     0.125000
RIVERDALE                         2025-03                     0.115607
```



Calendar month (pooled across years): count and median price — higher median ≈ pricier time of year in this sample

month_name	n	median_price
December	64	262500.0
July	55	250000.0
February	48	241500.0
September	62	240000.0
April	76	237500.0
March	78	235000.0
June	61	235000.0
October	68	231014.5
August	62	229000.0
January	61	223700.0
May	63	190000.0
November	55	179000.0

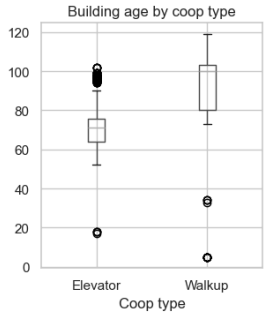
Top year-months by median price (min 10 sales):

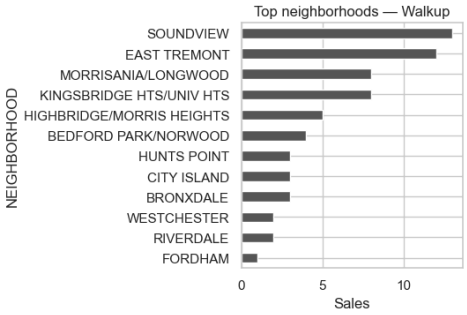
YEAR_MONTH	n	median_price	mean_price
2025-12	64	262500.0	273269.843750
2025-07	55	250000.0	242322.181818
2026-02	48	241500.0	246509.145833
2025-09	62	240000.0	245180.467742
2025-04	76	237500.0	235423.526316
2025-03	78	235000.0	251310.807692
2025-06	61	235000.0	247429.967213
2025-10	68	231014.5	243790.897059
2025-08	62	229000.0	262631.693548
2026-01	61	223700.0	259956.639344
2025-05	63	190000.0	234952.619048
2025-11	55	179000.0	218850.672727

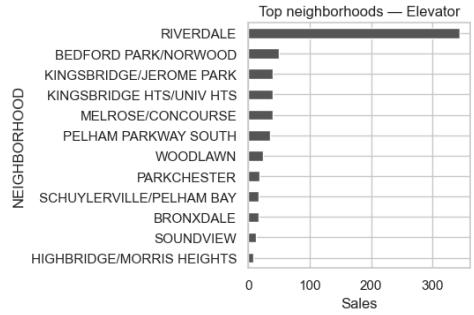
Highest median price by neighborhood × month (min 5 sales):

NEIGHBORHOOD	YEAR_MONTH	n	median
KINGSBRIDGE HTS/UNIV HTS	2025-08	7	450000.0
MELROSE/CONCOURSE	2025-12	5	399000.0
RIVERDALE	2025-11	17	340000.0
KINGSBRIDGE/JEROME PARK	2025-07	6	326803.0
RIVERDALE	2025-06	28	322500.0
RIVERDALE	2025-08	28	317750.0
RIVERDALE	2025-09	24	317500.0
RIVERDALE	2026-01	30	314750.0
RIVERDALE	2026-02	18	309000.0
RIVERDALE	2025-03	40	300350.0
RIVERDALE	2025-12	33	300000.0
RIVERDALE	2025-04	39	283500.0
RIVERDALE	2025-10	36	280000.0
RIVERDALE	2025-05	26	278750.0
RIVERDALE	2025-07	27	272000.0
BEDFORD PARK/NORWOOD	2025-07	5	270000.0
MELROSE/CONCOURSE	2025-03	5	269000.0
KINGSBRIDGE/JEROME PARK	2025-09	6	265000.0
MELROSE/CONCOURSE	2025-07	5	250000.0
KINGSBRIDGE/JEROME PARK	2025-03	6	227500.0

```
 KINGSBRIDGE/JEROME PARK    2025-04  6 217500.0
            PARKCHESTER    2025-03  6 175000.0
    BEDFORD PARK/NORWOOD    2026-01  8 174500.0
    PELHAM PARKWAY SOUTH    2025-08  5 165000.0
KINGSBRIDGE HTS/UNIV HTS    2026-01  5 165000.0

By quarter (top 12 by median):
           n  median_price
QUARTER
2026Q1   109      240000.0
2025Q1    78      235000.0
2025Q2   200      235000.0
2025Q3   179      230000.0
2025Q4   187      225000.0

Top 15 neighborhoods: sale counts and medians by coop type (columns = type)
— Bronx
                                n           median
COOP_TYPE                 Elevator Walkup  Elevator    Walkup
NEIGHBORHOOD
BEDFORD PARK/NORWOOD          49.0    4.0  149000.0  127500.0
BRONXDALE                     17.0    3.0  200000.0  100000.0
EAST TREMONT                   1.0   12.0  120000.0  148000.0
FORDHAM                        6.0    1.0  208500.0   85000.0
HIGHBRIDGE/MORRIS HEIGHTS      8.0    5.0  139500.0  130000.0
KINGSBRIDGE HTS/UNIV HTS      40.0    8.0  168000.0  149000.0
KINGSBRIDGE/JEROME PARK       40.0    NaN  230000.0       NaN
MELROSE/CONCOURSE             39.0    1.0  270000.0  225000.0
MORRISANIA/LONGWOOD            2.0    8.0  270000.0   79500.0
PARKCHESTER                   18.0    NaN  175000.0       NaN
PELHAM PARKWAY SOUTH          35.0    NaN  157000.0       NaN
RIVERDALE                    344.0    2.0  300000.0  195005.0
SCHUYLERVILLE/PELHAM BAY      17.0    NaN  168000.0       NaN
SOUNDVIEW                     12.0   13.0  365000.0  243000.0
WOODLAWN                      24.0    NaN  182500.0       NaN

Building age (years) by coop type:
           count       mean        std   min   25%    50%    75%    max
COOP_TYPE
Elevator   687.0  71.540029  11.660569  17.0  64.0   71.0   75.5  102.0
Walkup      66.0  81.818182  38.868975   5.0  80.0  100.0  103.0  119.0
```





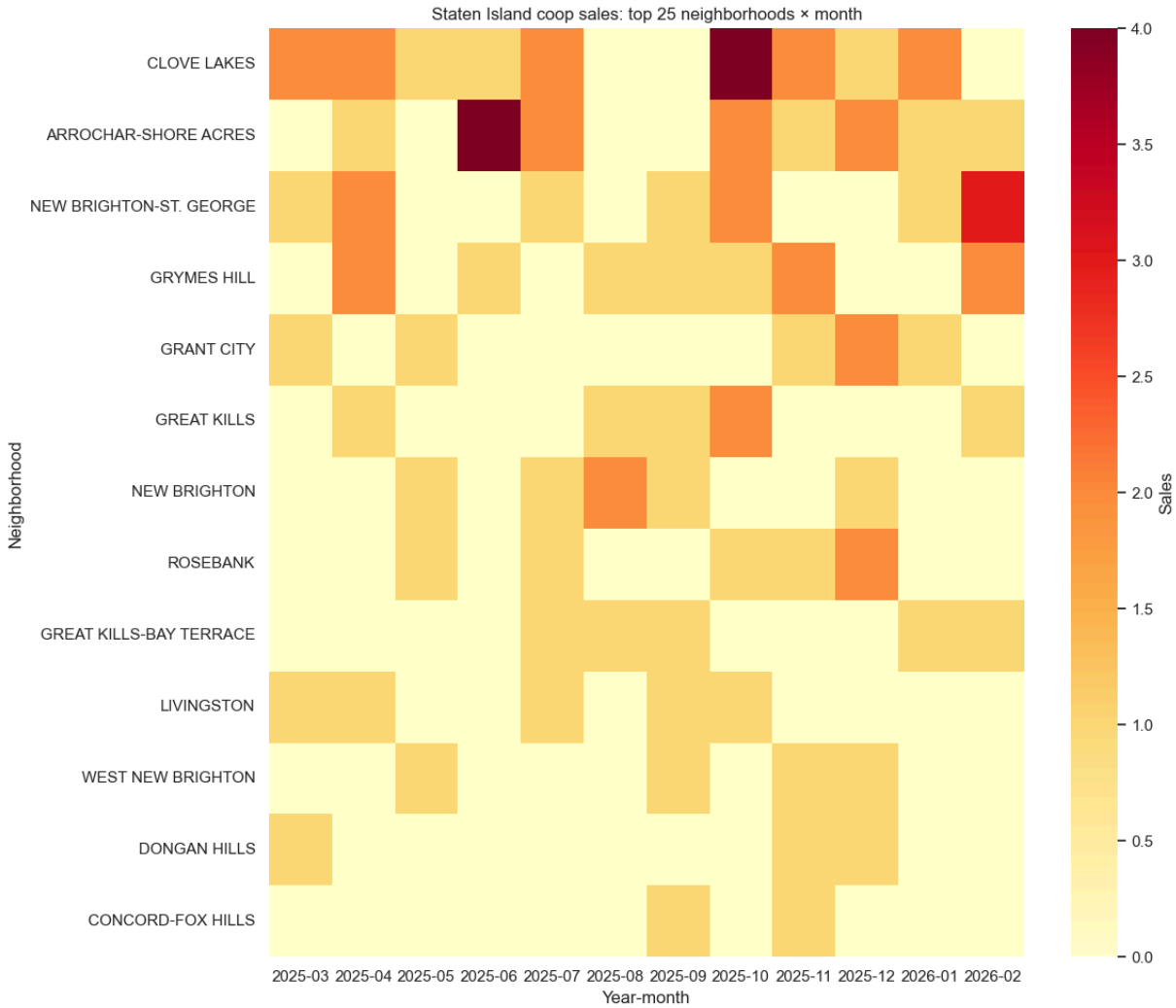


```
==========================================================================
 STATEN ISLAND COOPS
==========================================================================
Total coop sales (excludes CONDO COOPS and similar hybrids):
121

Building class categories:
<StringArray>
['10 COOPS - ELEVATOR APARTMENTS', '09 COOPS - WALKUP APARTMENTS']
Length: 2, dtype: str

Columns:
Index(['NEIGHBORHOOD', 'BUILDING CLASS CATEGORY', 'TAX CLASS AT PRESENT',
       'BLOCK', 'LOT', 'BUILDING CLASS AT PRESENT', 'ADDRESS', 'ZIP CODE',
       'RESIDENTIAL UNITS', 'COMMERCIAL UNITS', 'TOTAL UNITS',
       'LAND SQUARE FEET', 'GROSS SQUARE FEET', 'YEAR BUILT',
       'TAX CLASS AT TIME OF SALE', 'BUILDING CLASS AT TIME OF SALE',
       'SALE PRICE', 'SALE DATE'],
      dtype='str')

Head:
            NEIGHBORHOOD         BUILDING CLASS CATEGORY  \
480  ARROCHAR-SHORE ACRES  10 COOPS - ELEVATOR APARTMENTS
481  ARROCHAR-SHORE ACRES  10 COOPS - ELEVATOR APARTMENTS
482  ARROCHAR-SHORE ACRES  10 COOPS - ELEVATOR APARTMENTS
483  ARROCHAR-SHORE ACRES  10 COOPS - ELEVATOR APARTMENTS
484  ARROCHAR-SHORE ACRES  10 COOPS - ELEVATOR APARTMENTS

    TAX CLASS AT PRESENT  BLOCK  LOT BUILDING CLASS AT PRESENT  \
480                    2   2833   31                        D4
481                    2   2833   31                        D4
482                    2   2833   31                        D4
483                    2   2833   31                        D4
484                    2   2833   31                        D4

                 ADDRESS  ZIP CODE  RESIDENTIAL UNITS  COMMERCIAL UNITS  \
480  20 CLIFF STREET, 2H   10305.0                NaN               NaN
481  20 CLIFF STREET, 3K   10305.0                NaN               NaN
482  20 CLIFF STREET, 3R   10305.0                NaN               NaN
483  20 CLIFF STREET, 4F   10305.0                NaN               NaN
484  20 CLIFF STREET, 4M   10305.0                NaN               NaN

     TOTAL UNITS  LAND SQUARE FEET  GROSS SQUARE FEET  YEAR BUILT  \
480          NaN               NaN                NaN      1964.0
481          NaN               NaN                NaN      1964.0
482          NaN               NaN                NaN      1964.0
483          NaN               NaN                NaN      1964.0
484          NaN               NaN                NaN      1964.0

     TAX CLASS AT TIME OF SALE BUILDING CLASS AT TIME OF SALE  SALE PRICE  \
480                          2                             D4      270000
481                          2                             D4      208500
482                          2                             D4      150000
483                          2                             D4      262500
484                          2                             D4      340000
```

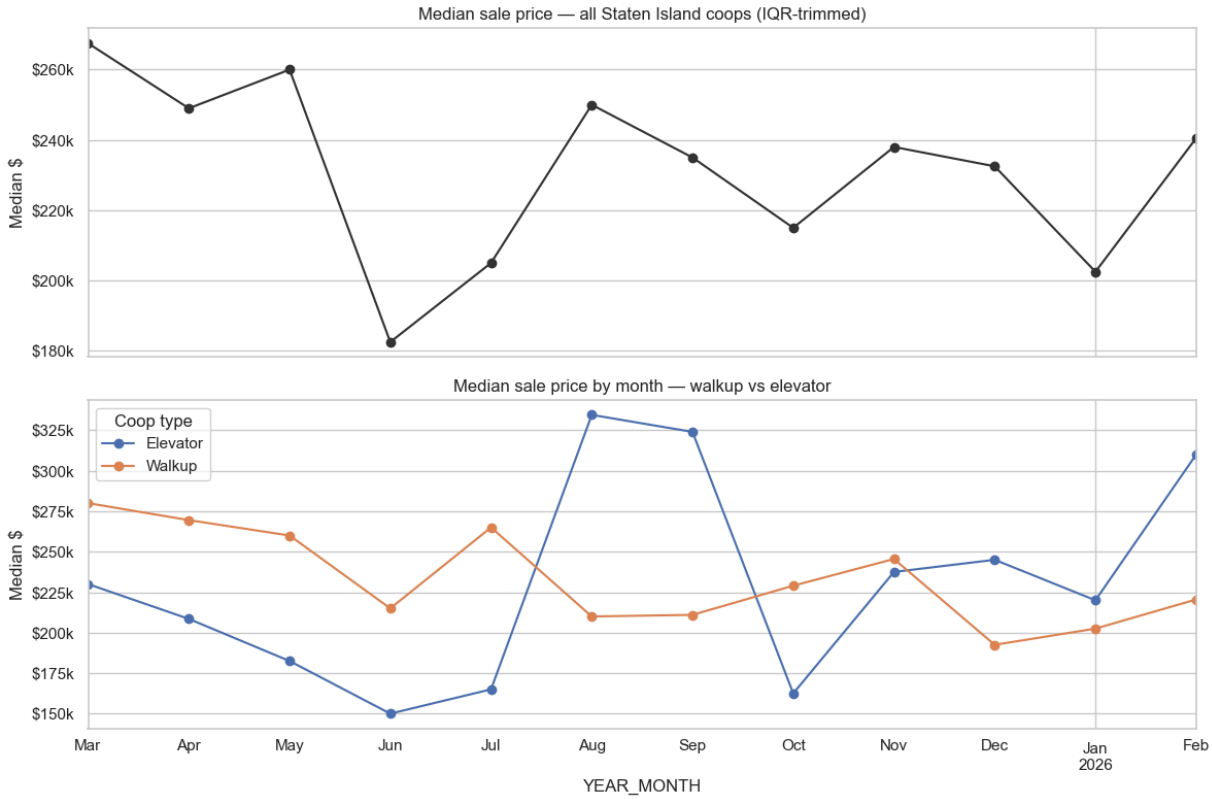
```
     SALE DATE
480 2025-10-15
481 2025-04-02
482 2025-07-01
483 2025-06-16
484 2025-12-15

Entirely null columns (dropped): ['RESIDENTIAL UNITS', 'COMMERCIAL UNITS',
'TOTAL UNITS', 'LAND SQUARE FEET', 'GROSS SQUARE FEET']
             NEIGHBORHOOD         BUILDING CLASS CATEGORY  \
480  ARROCHAR-SHORE ACRES  10 COOPS - ELEVATOR APARTMENTS
481  ARROCHAR-SHORE ACRES  10 COOPS - ELEVATOR APARTMENTS
482  ARROCHAR-SHORE ACRES  10 COOPS - ELEVATOR APARTMENTS
483  ARROCHAR-SHORE ACRES  10 COOPS - ELEVATOR APARTMENTS
484  ARROCHAR-SHORE ACRES  10 COOPS - ELEVATOR APARTMENTS

    TAX CLASS AT PRESENT  BLOCK  LOT BUILDING CLASS AT PRESENT  \
480                    2   2833   31                        D4
481                    2   2833   31                        D4
482                    2   2833   31                        D4
483                    2   2833   31                        D4
484                    2   2833   31                        D4

                 ADDRESS  ZIP CODE  YEAR BUILT  TAX CLASS AT TIME OF SALE  \
480  20 CLIFF STREET, 2H   10305.0      1964.0                           2
481  20 CLIFF STREET, 3K   10305.0      1964.0                           2
482  20 CLIFF STREET, 3R   10305.0      1964.0                           2
483  20 CLIFF STREET, 4F   10305.0      1964.0                           2
484  20 CLIFF STREET, 4M   10305.0      1964.0                           2

    BUILDING CLASS AT TIME OF SALE  SALE PRICE  SALE DATE
480                             D4      270000 2025-10-15
481                             D4      208500 2025-04-02
482                             D4      150000 2025-07-01
483                             D4      262500 2025-06-16
484                             D4      340000 2025-12-15

Fraction null per column:
NEIGHBORHOOD: 0.0
BUILDING CLASS CATEGORY: 0.0
TAX CLASS AT PRESENT: 0.0
BLOCK: 0.0
LOT: 0.0
BUILDING CLASS AT PRESENT: 0.0
ADDRESS: 0.0
ZIP CODE: 0.0
YEAR BUILT: 0.0
TAX CLASS AT TIME OF SALE: 0.0
BUILDING CLASS AT TIME OF SALE: 0.0
SALE PRICE: 0.0
SALE DATE: 0.0

After feature tweaks:
             NEIGHBORHOOD TAX CLASS AT PRESENT  BLOCK  LOT  \
480  ARROCHAR-SHORE ACRES                    2   2833   31
481  ARROCHAR-SHORE ACRES                    2   2833   31
```

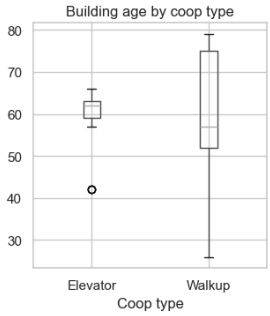
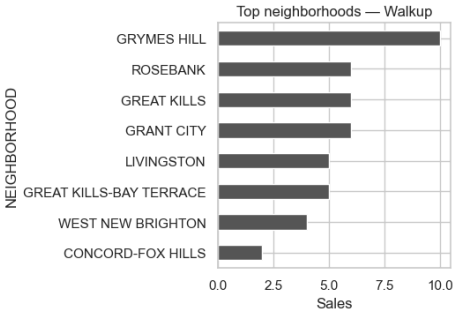
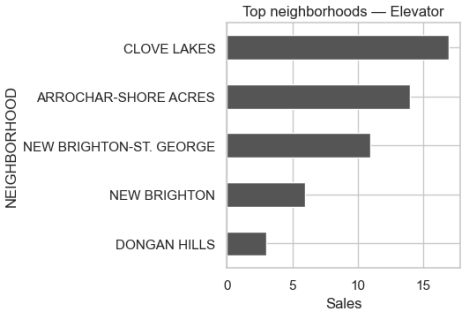
```
482  ARROCHAR-SHORE ACRES                    2   2833   31
483  ARROCHAR-SHORE ACRES                    2   2833   31
484  ARROCHAR-SHORE ACRES                    2   2833   31

    BUILDING CLASS AT PRESENT  ZIP CODE  YEAR BUILT  \
480                        D4   10305.0      1964.0
481                        D4   10305.0      1964.0
482                        D4   10305.0      1964.0
483                        D4   10305.0      1964.0
484                        D4   10305.0      1964.0

    BUILDING CLASS AT TIME OF SALE  SALE PRICE  SALE DATE  AGE OF THE BUILDI
NG
480                             D4      270000 2025-10-15                 6
2.0
481                             D4      208500 2025-04-02                 6
2.0
482                             D4      150000 2025-07-01                 6
2.0
483                             D4      262500 2025-06-16                 6
2.0
484                             D4      340000 2025-12-15                 6
2.0

Fraction above upper IQR fence: 0.21487603305785125
            NEIGHBORHOOD TAX CLASS AT PRESENT  BLOCK  LOT  \
480  ARROCHAR-SHORE ACRES                    2   2833   31
481  ARROCHAR-SHORE ACRES                    2   2833   31
482  ARROCHAR-SHORE ACRES                    2   2833   31
483  ARROCHAR-SHORE ACRES                    2   2833   31
484  ARROCHAR-SHORE ACRES                    2   2833   31

    BUILDING CLASS AT PRESENT  ZIP CODE  YEAR BUILT  \
480                        D4   10305.0      1964.0
481                        D4   10305.0      1964.0
482                        D4   10305.0      1964.0
483                        D4   10305.0      1964.0
484                        D4   10305.0      1964.0

    BUILDING CLASS AT TIME OF SALE  SALE PRICE  SALE DATE  AGE OF THE BUILDI
NG
480                             D4      270000 2025-10-15                 6
2.0
481                             D4      208500 2025-04-02                 6
2.0
482                             D4      150000 2025-07-01                 6
2.0
483                             D4      262500 2025-06-16                 6
2.0
484                             D4      340000 2025-12-15                 6
2.0

COOP_TYPE counts:
COOP_TYPE
Elevator    51
Walkup      44
```

```
Name: count, dtype: int64

Rows after upper IQR trim: 95
Sale date range: 2025-03-10 – 2026-02-20
Neighborhoods: 13
```




```
Peak year-month per neighborhood (top 25 by volume) and share of that hood's
sales in that month:
                        peak_year_month  share_of_nbhd_in_peak_month
NEIGHBORHOOD
CONCORD-FOX HILLS               2025-09                     0.500000
DONGAN HILLS                    2025-03                     0.333333
GRANT CITY                      2025-12                     0.333333
GREAT KILLS                     2025-10                     0.333333
NEW BRIGHTON                    2025-08                     0.333333
ROSEBANK                        2025-12                     0.333333
ARROCHAR-SHORE ACRES            2025-06                     0.285714
NEW BRIGHTON-ST. GEORGE         2026-02                     0.272727
WEST NEW BRIGHTON               2025-11                     0.250000
CLOVE LAKES                     2025-10                     0.235294
GREAT KILLS-BAY TERRACE         2026-02                     0.200000
GRYMES HILL                     2025-04                     0.200000
LIVINGSTON                      2025-09                     0.200000
```


Median sale price — all Staten Island coops (IQR-trimmed)
Median $
$260k
$240k
$220k
$200k
$180k
Median sale price by month — walkup vs elevator
Coop type
Elevator
Walkup
Median $
$325k
$300k
$275k
$250k
$225k
$200k
$175k
$150k
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Jan
2026
Feb
YEAR_MONTH

```
Calendar month (pooled across years): count and median price — higher median
≈ pricier time of year in this sample
             n  median_price
month_name
March        6      267500.0
May          5      260000.0
August       5      250000.0
April        9      249000.0
February     8      240500.0
November    10      238000.0
September    8      235000.0
December    10      232500.0
October     13      215000.0
July         9      205000.0
January      6      202500.0
June         6      182500.0

Top year-months by median price (min 10 sales):

             n  median_price     mean_price
YEAR_MONTH
2025-11     10      238000.0  235350.000000
2025-12     10      232500.0  234140.000000
2025-10     13      215000.0  187203.692308

Highest median price by neighborhood × month (min 5 sales):
Empty DataFrame
Columns: [NEIGHBORHOOD, YEAR_MONTH, n, median]
Index: []

By quarter (top 12 by median):
          n  median_price
QUARTER
2025Q1    6      267500.0
2025Q2   20      237000.0
2025Q3   22      235000.0
2026Q1   14      223000.0
2025Q4   33      220000.0

Top 15 neighborhoods: sale counts and medians by coop type (columns = type)
— Staten Island
                                n           median
COOP_TYPE               Elevator Walkup  Elevator    Walkup
NEIGHBORHOOD
ARROCHAR-SHORE ACRES        14.0    NaN  215000.0       NaN
CLOVE LAKES                 17.0    NaN  152500.0       NaN
CONCORD-FOX HILLS            NaN    2.0       NaN  212500.0
DONGAN HILLS                 3.0    NaN  330000.0       NaN
GRANT CITY                   NaN    6.0       NaN  237500.0
GREAT KILLS                  NaN    6.0       NaN  230000.0
GREAT KILLS-BAY TERRACE      NaN    5.0       NaN  205000.0
GRYMES HILL                  NaN   10.0       NaN  215000.0
LIVINGSTON                   NaN    5.0       NaN  275000.0
NEW BRIGHTON                 6.0    NaN  283500.0       NaN
NEW BRIGHTON-ST. GEORGE     11.0    NaN  265000.0       NaN
ROSEBANK                     NaN    6.0       NaN  265000.0
```

```
WEST NEW BRIGHTON            NaN    4.0       NaN  212500.0

Building age (years) by coop type:
           count       mean        std   min   25%   50%   75%   max
COOP_TYPE
Elevator    51.0  60.117647   5.042408  42.0  59.0  62.0  63.0  66.0
Walkup      44.0  61.636364  14.224268  26.0  52.0  57.0  75.0  79.0
```